

Warm-up 1

Find the eigenvalues of each of the following matrices by solving the equation $\det(A - \lambda I) = 0$.

$$\begin{pmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix} \quad \begin{pmatrix} 4 & 7 & 9 \\ 0 & 3 & -8 \\ 0 & 0 & 2 \end{pmatrix} \quad \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

During class 1

Give an example of a real 2×2 matrix for each of the four regions of a Venn diagram where the two categories are “invertible” and “diagonalizable”; or for a given region, explain why no such matrix exists.